PA194J and PA194D: two new 194-state PA consistency machines

Corpus comparison, zero-register jumps, a changed diagonal emitter, and PA-provable equivalence

Research report for the ProveIt PA project

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Abstract

A machine with fewer than 194 working states was not obtained. We provide the requested alternative: a materially different 194-state, two-symbol, single-tape machine, PA194J, with an ordinary mathematical proof of PA $\vdash [\neg\,\mathrm{Halt}(\mathsf{PA194J}, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathrm{PA})]$. The new machine replaces ten unconditional source jumps by decrements of registers proved zero at those locations. A finite inductive abstraction with 263,180 location/status pairs establishes exact equality of the parent and child source runs from their common seed. Three transition-row changes implement these substitutions, retaining the symmetric Shift controller and the existing 67-reader decoder.

The construction contains 194 working states and 388 transition rows. Its source has 3,646 primitive sites and 4,755 branch outcomes. A literal tape checker derives four generic list lemmas from the actual rows and checks the source outcomes for arbitrary natural register values, gap, and finite nonempty spare list. Exact blank initialization takes 60,062 transitions and reaches the complete expected tape. The new graph differs from every existing 194-state graph in the pinned corpus, even allowing state renaming, a different designated start, reflection, and global symbol exchange.

A second construction, PA194D, changes the logical emitter from $\Gamma(A)$ to $\Gamma(D)$, with a correspondingly changed selector in the strong proof-word compiler. A new allocation of 69 decoder readers admits nine simultaneous history folds, giving $134 + 69 - 9 = 194$ working states. Its exact blank startup takes 58,168 transitions. It is supplied as a separate literal table and construction, with its own calibration argument and physical checks.

The comparison covers 41 research packets at ProveIt revision `17f92105e2bc4317feae6c219cb2143437b482ff`: 36 filed packets and five newer incoming archives. The census recognizes 2,685 table occurrences, 620 byte forms, and 446 action-graph classes; rejected and diagnostic tables are identified separately. The proof transfers the supplied ordinary PA calibration through new source and physical preservation arguments. No new end-to-end Lean or Rocq theorem, iPRA equivalence, state lower bound, or unconditional nonhalting is claimed.

Contents

1 Result and precise machine convention **3**
1.1 Contribution and cost . . . 3

2 Comparison of the supplied designs **4**
2.1 Scope and reproducible census . . . 4
2.2 The eight existing 194-state graphs . . . 5
2.3 Larger families and transfer of ideas . . . 5

3 The new construction **6**
3.1 Registers, source instructions and the common seed . . . 6
3.2 Ten replacements, implemented by three rows . . . 6
3.3 Why a reachable invariant is essential . . . 7

4 An inductive proof of exact source-run preservation **7**

5 Binding the source to the literal binary table **8**
5.1 Complete symbolic frames . . . 8
5.2 Universal physical contracts . . . 9
5.3 Exact blank initialization . . . 9

6 The inherited PA consistency calibration **10**
6.1 Numerical formulas and commands . . . 10
6.2 A source halt produces a PA refutation . . . 11
6.3 A PA refutation produces a source halt . . . 11

7 PA194D: a different logical emitter with 194 states **12**
7.1 The exact logical change . . . 12
7.2 All-store command identities and the changed selector . . . 13
7.3 A strong proof compiler and both logical directions . . . 15
7.4 Exact source binding and the two zero tests . . . 16
7.5 Literal realization and the PA theorem . . . 17

8 Novelty within the pinned corpus **18**

9 Searches below 194 and their limits **18**
9.1 Two concrete failures worth preserving . . . 19
9.2 What transfers from ZFC and concise iPRA . . . 19

10 Reproduction and evidence boundaries **20**

11 Further research questions **20**

A Complete dense transition table **22**

B Complete dense PA194D transition table **24**

C All 41 research packets **26**

1 Result and precise machine convention

The principal file is `construction/machine/pa194j.tm`. The machine starts in working state 0, scanning cell 0 of a two-way infinite tape filled with zero. The alphabet is $\{0, 1\}$. Working states are $0, \ldots, 193$; label 194 is the terminal halt control, has no outgoing rows, and is excluded from the state count. Each transition writes a bit, moves exactly one cell left or right, and enters its target state. Every working state has exactly two rows. The five-column format is

`stateread_bitwrite_bitL_or_Rnext_state.`

All 194 working controls are reachable in the pointed transition graph. This finite graph fact does not assert that the blank execution visits every control.

The authoritative dense table has SHA-256

`7c498ced0eb0dffed77ae2f68f4d80c98edfd8f7763ec1be68aa8cd7ea80eacd.`

Its native-label presentation starts at 74 and halts at 254, with SHA-256

`31abf450ba25066c8ac3aad6e7eedb59386b770bb4917f006800145ebb622218.`

The supplied label map conjugates every transition exactly. The entire dense table is printed in Appendix A.

Theorem 1.1 (Main result)**.** *For the literal table specified above, with* H_{PA194J} *the usual arithmetic sentence asserting a finite halting trace from its blank start,*

$$\mathrm{PA} \vdash (H_{\mathsf{PA194J}} \leftrightarrow \neg\,\mathrm{Con}(\mathrm{PA})), \qquad \mathrm{PA} \vdash (\neg H_{\mathsf{PA194J}} \leftrightarrow \mathrm{Con}(\mathrm{PA})).$$

Here the proof is an ordinary mathematical proof formalizable in PA, with executable finite and symbolic certificates. Its logical calibration is inherited from the supplied PA194/PA195 developments [2, 3] and retained in `references/logical_calibration/`. The machine-specific work in this packet is the new source invariant, exact table construction, independent physical simulation, and exact initialization. The repository's unfinished iPRA endpoints and the absence of a new kernel export remain distinct from this ordinary proof; see Section 6.

1.1 Contribution and cost

The immediate parent is the supplied PA194CS machine, which already uses a symmetric destructive comparison in Shift. PA194J specializes its control transfers using *global source invariants*. At eight sites register A is zero; at the scheduler site register X is zero; at the final AX return register D is zero. At these ten sites a decrement can implement precisely the former jump. These substitutions create new pure right-moving readers in the physical decoder.

The count stays

$$\underbrace{135}_{\text{fixed part before the CS fold}} + \underbrace{67}_{\text{raw AX readers}} - \underbrace{8}_{\text{seven AX folds and the CS fold}} = 194.$$

Equivalently, after the controller fold there are 134 fixed controls and 60 AX controls. The new construction improves neither this count nor the startup time: initialization grows from the parent's 57,768

transitions to 60,062. Its value as the requested fallback is a different verified physical implementation with new decoder interfaces and an explicit global invariant available for further state reductions.

This is not merely a different numbering of an old table. The action graphs are distinct under the broad finite equivalence tests in Section 8. Two changed rows are exercised 3 and 21 times, respectively, during the actual blank initialization. At source level the program now has ten conditional instructions where the parent had jumps; the proof explains why their zero branches always apply.

2 Comparison of the supplied designs

2.1 Scope and reproducible census

All comparisons are pinned to the revision on the title page [1]. The `PA/docs` ledger contains 36 delivered packets. Five later incoming archives are absent from that ledger and were read separately:

- `pa194C_conpa_2026-11-13(2).zip`;
- `pa194_new_designs_2026-11-13.zip`;
- `pa194c_conpa.zip`;
- `pa194c_conpa_2026-11-13(1).zip`;
- `pa194c_conpa_2026-11-13.zip`.

The byte identities of these five archives were checked against their Git blob identities. All five repeat the same principal PA194C table. Some additionally contain PA194CS, PA194CZ, or an eight-fold cached search input. Treating every delivery as a new machine would therefore misdescribe the current frontier.

The recognizer reads complete numeric binary tables in `.tm`, `.txt`, `.tsv`, and `.csv` files, including recursively nested archives. It checks two rows per control, unique source/read pairs, binary symbols, L/R moves, and one undefined halt target. The filed collection replaces earlier archive wrappers, so occurrence counts are not directly comparable to an incoming-only traversal. To check coverage, every one of the 606 byte identities in the prior full census was recovered and its action-graph identity recomputed. The new census adds 14 byte forms.

Quantity	Fresh result
Research packets	41
Filename-selected file occurrences inspected	4,570
Recognized complete numeric-table occurrences	2,685
Byte-distinct complete tables	620
Root-free action-graph classes	446
194-state occurrences / byte forms / graph classes	95 / 15 / 8

These are artifact counts, not numbers of correct consistency machines. Two disconnected synthetic controls with six and nine states are excluded from root-free classification; they are the only recognized sub-194 tables and are not proposed PA machines. The recognizer does not recover arbitrary transition systems embedded only in program source or JSON. The proposed frontier tables are present literally. The full member ledger, frozen recognized tables, and comparison code are included.

2.2 The eight existing 194-state graphs

Design	Sites	Outcomes	Startup	Distinguishing construction
PA194-67	3,673	4,780	58,154	67 readers, seven compatible folds.
PA194-67-AD	3,673	4,780	58,122	Physical exchange of A/D register roles.
PA194-68	3,673	4,780	58,154	68 readers, eight compatible folds.
PA194C	3,649	4,749	57,922	Cached MP conclusion and cached-head Shift.
PA194CS	3,646	4,745	57,768	Symmetric Shift removes a restoration loop.
PA194CZ	3,649	4,750	58,168	One zero-register jump at source site 26.
PA194-altC	3,649	4,749	57,922	Cached controller with the eight-fold decoder.
Rejected scanner	—	—	—	Too-short scanner misreads a legal source path.

The old runtime and startup numbers in this table are those recorded by the supplied constructions; this task freshly checked table identities, source comparisons and novelty, and freshly verified the delivered new machine. It did not rerun every historical physical proof.

The complete ordinal instruction graphs of PA194-67 and PA194-68 are equal. Their seven and eight folds belong to different reader allocations; they cannot be accumulated as fifteen independent savings. PA194C has exactly the earlier PA195C source. PA194CS agrees with PA194C through ordinal 3643 and replaces its last five-site Shift suffix by $[\mathrm{inc}\,5; \mathrm{jump}\,0]$. Its extra reader is balanced by the contextual identification $41 \to 330$.

PA194-altC is easy to miss: it is a checked search input inside `pa194c_conpa.zip`, rather than the packet's headline result. It belongs in a novelty test nonetheless. The rejected scanner graph also belongs in a byte/graph census, but not in a list of valid constructions. Its source address at ordinal 1003 has seven consecutive zeroes, which the shortened scanner mistakes for header padding. A local fold proof alone does not prevent that global error.

2.3 Larger families and transfer of ideas

The U208 packets exploit a finite-history identification of controls 172 and 211 in a U209 ancestor. Many deliveries are alternate presentations of the same graph. The relevant controls have already disappeared in the PA196 development, so their saving cannot simply be applied once more to PA194. James Chang's PA196 source has 3,812 primitive sites; the later logical emitters and their layout searches account for the transition to the current frontier.

The main 195-state families illustrate different routes to the same count. Original PA195 uses a 191-token bundled diagonal emitter and 3,700 sites. PA195S separates the diagonal clause, using 190 tokens and 3,673 sites, with $134 + 67 - 6$ controls. PA195C adds the cached controller, reducing the source to 3,649 sites. PA195DC combines the bundled emitter with that controller and has 3,676 sites. M195S8 achieves eight folds but pays for an additional scanner control. These tradeoffs explain why a source simplification or a larger fold count alone is insufficient.

NEG18 is another useful example. Its arithmetic identity shortens encoded negation from 47 primitive sites to 18. In the separate-head source, replacing eight occurrences reduces 3,673 sites to 3,441, yet its published physical implementation has 197 states. Reader sharing, legal return paths and compatible local folds dominate the final cost. The new searches in this packet included mixed negation spellings on the newer grammar seeds; none supplied a certified sub-194 table.

Appendix C gives an archive-by-archive comparison of all 41 packets. `comparison/archive_ledger.csv` also records exact locators, hashes and evidence qualifications.

3 The new construction

3.1 Registers, source instructions and the common seed

The nine logical registers, numbered from zero, are

$$(N, T, W, A, D, H, X, Y, L).$$

They are stored at physical unary positions $(6, 2, 0, 4, 3, 7, 8, 5, 1)$, respectively. A source instruction is one of $\mathrm{inc}(r, u)$, $\mathrm{dec}(r, z, p)$, $\mathrm{jump}(u)$, or halt. A decrement takes successor z and leaves the registers unchanged when register r is zero; otherwise it subtracts one and takes successor p.

The clean source seed shared by PA194CS and PA194J is

$$c_* = (25, (2, 0, 0, 0, 0, 0, 0, 0, 0)).$$

The source path of site 25 is `1000000`. The physical clean frame has native control 84, head at -2, header distance 48, the nine unary registers above, and 26 spare blocks of value one. Both machines are initialized from a literal blank tape; their different finite startup traces are verified separately.

3.2 Ten replacements, implemented by three rows

Site	Zero register	Old jump / new zero target	New positive target	Masks
3	A (3)	5	4	32
9	A (3)	11	10	16
26	X (6)	34	27	32
3526	A (3)	3528	3527	32
3532	A (3)	3535	3533	16
3544	A (3)	3546	3545	16
3550	A (3)	3553	3551	8
3553	D (4)	3637	3554	8
3627	A (3)	3629	3628	16
3633	A (3)	3637	3634	8

"Masks" counts the statuses occurring at that site in the finite inductive abstraction defined below, not the number of possible natural register vectors. All source paths are retained. Every instruction outside these ten sites is identical, including its complete successor list. The new source has 1,109 decrement sites, hence $3646 + 1109 = 4755$ primitive outcomes.

In native labels, the complete difference from PA194CS is

Control	Read	Parent action	PA194J action
27	1	$0\,L\,40$	$1\,R\,21$
103	1	$0\,L\,302$	$1\,R\,17$
2066	1	$0\,L\,302$	$1\,R\,144$

Controls 21, 17 and 144 enter the respective decrement interfaces. The last two changes provide reader pairs $(108, 17)$ at control 103 and $(221, 144)$ at control 2066. The first provides $(39, 21)$ at control 27. A pure reader preserves the scanned bit and moves right; these pairs are useful as possible helper interfaces in subsequent history-based folds. Their availability has not yet yielded a saved state.

The eight shared replacements arise from the source-tree definition `node1=(node0,breakTo1)`, changed to `node1=(node0,dec3)`. The scheduler definition changes `node22=(dec8,breakTo1)` to `(dec8,dec6)`. The unique final AX `breakTo1` becomes `dec4`. The builder regenerates the whole 3,646-site source from these tree edits; it does not merely patch a supplied successor list.

3.3 Why a reachable invariant is essential

The replacement at site 9 has positive successor 10, a halt site. Thus the two source programs are not interchangeable on arbitrary register inputs. The proof must establish that the positive branch is unreachable from the common seed. It does so for all ten replacements simultaneously. In contrast, the *physical simulation* theorem checks every branch for arbitrary registers, including the positive branches. These two different scopes fit together: the source invariant selects the actual branches, while the physical theorem implements the complete new source.

4 An inductive proof of exact source-run preservation

For a natural register vector $r = (r_0, \ldots, r_8)$, define its positivity mask

$$\alpha(r) = \sum_{j=0}^{8} 2^j [\, r_j > 0 \,] \in \{0, \ldots, 511\}.$$

Write $m \setminus j$ for clearing bit j and $m \cup j$ for setting it. Abstract states are pairs (i, m) of a source ordinal and a mask. For an instruction at site i, define the abstract successors by

Instruction	Abstract successors
$\mathrm{jump}(u)$	$\{(u, m)\}$
$\mathrm{inc}(j, u)$	$\{(u, m \cup j)\}$
$\mathrm{dec}(j, z, p),\ m_j = 0$	$\{(z, m)\}$
$\mathrm{dec}(j, z, p),\ m_j = 1$	$\{(p, m), (p, m \setminus j)\}$
halt	$\varnothing$.

The two positive-decrement possibilities reflect the distinction between an input equal to one and an input greater than one. Discarding that distinction is a deliberate overapproximation.

Lemma 4.1 (Sound abstraction)**.** *Every concrete source step* $(i, r) \to (i', r')$ *projects to an allowed abstract step* $(i, \alpha(r)) \to (i', \alpha(r'))$.

Proof. Jumps preserve the vector. An increment makes its component positive. A decrement of zero preserves the vector and takes the zero target. If the component is positive, subtracting one either makes it zero or leaves it positive, with every other bit unchanged. These cases are exhaustive over the natural numbers. □

Let P be the parent PA194CS source and P' the PA194J source. Compute the least abstract closure R of $(25, 1)$ under P. The included builder, a separate finite checker, and an independent review implementation reproduce the following finite facts:

1. $|R| = 263180$, and $(25, 1) \in R$.
2. The abstract successors of every element of R lie in R.

3. At every element of R, the P and P' abstract successor sets agree exactly. Recomputing the least closure for P' gives the same R.
4. At each changed site, the chosen register bit is zero in every mask in R at that site.

The canonical sorted-pair certificate has SHA-256

`9191fff6d156190b0eaa97f93dd5d10a23fbeac2d6505c0a269108e91c86f483`.

The certificate is a finite object, not a bounded simulation of a source run. It says nothing about a bound of 263,180 execution steps; the invariant applies at every step, for arbitrarily large register values.

Lemma 4.2 (Invariant and zero tests)**.** *Every source configuration reachable from c_*, in either program, projects into R. Consequently every inserted decrement takes its zero branch and leaves every register unchanged.*

Proof. The seed projects to $(25, 1)$. Induct on the length of a concrete source run, using sound abstraction and finite closure for the inductive step. At a modified site, the corresponding bit in the projected mask is zero, which is equivalent to the natural register being zero. □

Theorem 4.3 (Exact source identity from the seed)**.** *For every finite number of source steps, P and P', started at c_*, have identical source ordinals and identical nine-register vectors, until and including any halt. In particular,*

$$\mathrm{PA} \vdash \big(\mathrm{Halt}(P, c_*) \leftrightarrow \mathrm{Halt}(P', c_*)\big).$$

Proof. Induct on steps while using the preceding invariant. At an unchanged site the concrete instructions agree. At a changed site the parent jumps to u, while the child decrements a register equal to zero and takes precisely u, preserving the vector. The halt sites themselves are unchanged. Thus the concrete states agree at the next step.

All definitions involved are primitive recursive. The finite closure claims can be expressed using a fixed coded finite set and checked by bounded computations in PA. The abstraction lemma is elementary arithmetic, and the remaining argument is ordinary induction on run length. The proof therefore establishes the displayed internal PA statement, not only standard-model agreement. □

The source proof depends on the seed. It is neither an all-tape equivalence between the literal machines nor an arbitrary-register equivalence between the source programs. No such stronger statement is needed for the blank-start theorem.

5 Binding the source to the literal binary table

5.1 Complete symbolic frames

A source frame represents a source path, zero padding, the header delimiter, nine unary registers and a finite nonempty list of spare unary blocks. The block for a value v is $B(v) = 0\,1^{v+1}$. The checker also uses $C(v) = 1\,0\,1^{v}$ during sweeps. It represents the complete tape, including the zero exterior, rather than a finite observed window. The base header distance is 48; its extra gap is an arbitrary natural. The primary values, first spare value and remaining spare-list width are independent symbolic quantities.

The source map is resolved twice, by an arithmetic path resolver and an interval resolver. Both give exactly the generated instruction list. This matters because equality of a flattened token word alone need not preserve the targets of structured `breakTo` and `continueTo` instructions.

5.2 Universal physical contracts

The independent checker derives four generic list-sweep lemmas from the literal transition rows, for arbitrary values and arbitrary exterior contexts. Finite-list induction extends the one-block calculations to all finite spare lists. Literal steps, monotone unary-loop summaries, these list lemmas, and exact word normalization then prove the source contracts.

For each increment or jump there is one outcome; for each decrement, there are separate zero and positive outcomes. A nonhalting outcome must reach the precise target source frame with the required register effect and spare-list update. A halt outcome must reach the terminal control. Every duration is a positive affine expression in the symbolic parameters. Consequently the simulation is productive: a source step cannot be represented by a zero-length physical loop.

Check	Result
Working states / transition rows	194 / 388
Primitive source sites / outcomes	3,646 / 4,755
Fresh symbolic rule applications	1,103,582
Maximum symbolic steps per case	403
Minimum actual-duration constant	3
Generic list lemmas derived from literal rows	4
Arithmetic-box oracles used	0
Independent source resolvers in agreement	2
Exact blank startup transitions	60,062

The complete physical replay returned **PASS**. The canonical contract list has SHA-256

`6c49226552b828b9d0325f358df908dc83995b0e08e851b38900db3c08124b9e`.

The checker imports no allocator or abstract arithmetic-box oracle. It is still ordinary executable mathematical evidence, not a proof assistant kernel. The inherited symbolic engine, its normalization identities and finite-list proof rules are exposed as readable source under `construction/reference/`, with their provenance and licenses.

5.3 Exact blank initialization

A separate bytearray simulator executes precisely 60,062 literal transitions from native control 74 on the all-zero tape. It obtains native control 84, head -2, and the frame of c_*:

$$\text{path} = 1000000, \quad \text{header distance} = 48, \quad (N, T, W, A, D, H, X, Y, L) = (2, 0, 0, 0, 0, 0, 0, 0, 0).$$

There are 26 spare blocks of value one and exactly 66 one-cells in the whole tape. Every cell outside the explicit frame is zero. A set-based simulator independently reproduces the same finite-support tape. Thus initialization is a complete configuration equality, not a check of only the register values or a small neighborhood of the head.

Proposition 5.1 (Physical halting equivalence)**.** *For PA194J, PA proves*

$$H_{\mathsf{PA194J}} \leftrightarrow \mathrm{Halt}(P', c_*).$$

Proof. The finite initialization reaches the represented source seed without halting. Concatenate the terminating positive physical simulations of source instructions. A source halt gives a finite physical halting trace. Conversely, suppose a physical halt occurs after t further transitions. If no source halt occurred, the first $t+1$ source instructions would have terminating nonhalting simulations, each of positive duration. Their concatenation extends beyond time t, contradicting the alleged halt. Intermediate tape configurations need not have a canonical source decoding for this argument. The explicit finite traces, unary and finite-list inductions, and induction on concatenated source steps are all formalizable in PA. □

The same established interface for the supplied PA194CS source [3] yields the corresponding parent statement. Combining with Section 4 gives PA-provable blank halting equivalence between PA194J and PA194CS.

6 The inherited PA consistency calibration

It would be insufficient to infer PA-provability merely from a claim that both machines do not halt in the standard model. The supplied calibration instead provides syntactic transformations in both directions. The relevant proof texts are retained in `references/logical_calibration/`, with the cached-command correction and the ten-entry selector adjustments. The inherited symmetric-guard proof is retained separately as `proof_symmetric.tex`; it establishes the precise cached command interface of PA194CS [3].

6.1 Numerical formulas and commands

Let

$$p(a,b) = \frac{(a+b)(a+b+1)}{2} + b, \quad I(A,B) = 4p(A,B), \quad M(a,b) = 4p(a,b)+1,$$
$$U(a,B) = 4p(B,a)+2, \qquad E(a,b) = 4p(a,b)+3.$$

The readings are implication, membership, universal quantification, and equality. Code zero is truth; code one, $M(0,0)$, is contradiction. Negation is $I(B,1)$. Formula-code bounds make the diagonal clause $\Gamma(B) = I(B, U(B,B))$ sound: variable x_B is absent from the formula coded by B. The current emitter pushes this clause separately from the guarded pointed-separation gate.

In the cached command system, stores are $\langle t, w, d, h\rangle$. Shift halts exactly when $h = 1$; otherwise it maps the store to $\langle h, \mathrm{tl}(w), d, \mathrm{hd}(w)\rangle$. For $y = d + x$, modus ponens maps it to $\langle 0, w, 0, y\rangle$ if $I(t,y) = h$, and to $\langle 0, w, y, 0\rangle$ otherwise. AX supplies ten pushed entries, counting the saved top, and clears the counter and cache.

PA194CS implements the same command behavior through its symmetric destructive equality routine. PA194J has exactly the same seeded primitive execution as PA194CS, so it needs no changed command calculus, selector, axiom schema or proof-word encoding.

The ten-entry command identities use

$$\begin{aligned}\mathsf{Push}_{10} &= \mathsf{AX}_{0,0,0}; S^{9}; \mathsf{MP}_{0},\\ \mathsf{Pop}_{10} &= \mathsf{AX}_{0,0,0}; S^{11},\\ \mathsf{Discard}_{10} &= \mathsf{MP}_{1}; \mathsf{AX}_{1,0,0}; S^{10}; \mathsf{MP}_{0}.\end{aligned}$$

The supplied command audit proves the full store effects, treats early halts explicitly, and uses the total Cantor projection identities even at $p(0,0) = 0$. In particular, the old nine-entry exponents cannot be used unchanged. This interface has already been corrected in the parent work and is preserved here.

6.2 A source halt produces a PA refutation

Write $\mathrm{Bit}(s, z)$ for "bit z of the finite set-code s is one." The interpretation sends numerical membership to finite binary membership $\mathrm{Bit}(x_a, x_b)$, equality to arithmetic equality, and U to universal quantification. PA proves that an element of the finite set coded by s is smaller than s, so it refutes the interpretation of code one. Pointed separation is interpreted using finite binary comprehension. For each arithmetic formula $\theta(z, \vec{u})$, the induction proof of

$$\forall n\, \exists s < 2^n\, \forall z < n\, (\mathrm{Bit}(s, z) \leftrightarrow \theta(z, \vec{u}))$$

is obtained by a fixed proof-generating template. Substitution of a formula's syntax into this template is primitive recursive, and PA verifies its output as a proof. This is a statement about proof codes, not a uniform truth predicate or a reflection principle.

Annotate the visible formula registers and canonical stack heads with PA proofs of their interpretations. AX, MP and Shift preserve those annotations. The zero-pair alias is handled by the truth annotation, and parameter collisions in pointed separation use the corresponding separate comprehension case. At a halt, the cache is code one, hence the annotations yield a PA contradiction. The supplied argument is a syntactic finite-trace transformation; it gives the PA-provable implication from source halting to $\neg\,\mathrm{Con}(\mathrm{PA})$.

6.3 A PA refutation produces a source halt

The reverse translation recovers the required equality and membership interfaces from the diagonal calculus. Reverse membership rewriting and neighborhood equality use code freshness and diagonal vacuity; the full proof is retained in `diagonal_calculus.tex` and `original_pa195_calibration.tex`. Fixed proof templates turn the bundled diagonal gate into the separate clause and gate, and back. The correctly indexed ten-entry command compiler then translates a refutation into a halting command word.

The strong form of this compiler is essential: its final word succeeds from every incoming store, or an earlier halt has already occurred. The scheduler retains stores between candidates, so a compiler requiring a fresh empty store at each candidate would not suffice. The supplied universal word begins with appropriate cleanup and handles every preserved stack tail. The exhaustive scheduler eventually reaches that word. All decoding, finite proof construction and simulation witnesses are primitive recursive, with correctness proved in PA.

Consequently the parent source satisfies

$$\mathrm{PA} \vdash \big(\mathrm{Halt}(P, c_*) \leftrightarrow \neg\,\mathrm{Con}(\mathrm{PA})\big).$$

The new theorem follows by the two equivalences already proved:

$$H_{\mathsf{PA194J}} \longleftrightarrow \mathrm{Halt}(P', c_*) \longleftrightarrow \mathrm{Halt}(P, c_*) \longleftrightarrow \neg\,\mathrm{Con}(\mathrm{PA}).$$

This composition requires no assumption that PA is consistent. If PA is consistent, actual nonhalting follows. No new end-to-end Lean/Rocq elaboration of the composed theorem is asserted.

7 PA194D: a different logical emitter with 194 states

There is a second 194-state construction, denoted PA194D, which changes the numerical axiom command itself. The separate diagonal clause in the earlier machines is $\Gamma(A) = A \to U(A, A)$. PA194D supplies $\Gamma(D) = D \to U(D, D)$ instead, while preserving the parameter choices in all six other heads and in the guarded separation gate. The resulting proof system is the same numerical calculus, but its command interface and its physical reader allocation are different. Its state count is

$$134 \text{ fixed controls} + 69 \text{ raw readers} - 9 \text{ compatible folds} = 194.$$

The command controller is the cached controller with the scheduler's known-zero decrement, as in the supplied PA194CZ construction [3, 4]. In addition, the final AX return uses a decrement of the cleared register D. The argument below separates the logical change from these two reachable source specializations.

The dense binary table has working controls $0, \ldots, 193$, blank start control 0, and terminal halt label 194. Its SHA-256 is

`4226b7702de554df4b52d10239b2bd7ad79ca90bd8faf2bd256552fe0c7fe81b`.

The native-label table starts at 74, halts at 254, and has SHA-256

`bc8ab6c58badd137535938c9810e39d19152c49dd2eb26ef4ea07dbcf3157231`.

The table uses the same two-way infinite binary tape convention as PA194J. It has 388 defined transition rows. The authoritative file is `logical_emitter/machine/pa194d.tm`; its complete contents are printed in Appendix B.

7.1 The exact logical change

Use Cantor pairing p, implication I, membership M, universal quantification U, and equality E with the numerical definitions in Section 6. In particular, $U(i, B) = 4p(B, i) + 2$, $\neg B = I(B, 1)$, and $\exists x_i B = \neg U(i, \neg B)$. Write $[a; w] = p(a, w)$, with the leftmost entry of a displayed list at its top. The six retained heads, in their original push order, are

$$\begin{aligned}
h_1(D, H) &= (\exists x_D U(D, H)) \to H,\\
h_2(D, A, H) &= U(D, U(A, H)) \to U(A, U(D, H)),\\
h_3(D, A, H) &= M(H, D) \to (E(H, A) \to M(A, D)),\\
h_4(D, A, H) &= E(H, D) \to (M(A, H) \to M(A, D)),\\
h_5(D, H) &= \exists x_D E(H, D),\\
h_6(D, A, H) &= U(D, A) \to (U(D, A \to H) \to U(D, H)).
\end{aligned}$$

The dominated propositional axiom is

$$\rho_8(D, A, H, u) = ((H \to u) \to D) \to ((D \to H) \to (A \to H)), \qquad u \le D.$$

Let $G(a, b, c)$ be the retained pointed-separation formula from the supplied separate-head calculus, and put

$$K(a, b, c) = \begin{cases} G(a, b, c), & c \neq a \text{ and } c \neq b, \\ 0, & \text{otherwise.} \end{cases}$$

For precision, its construction is

$$\begin{aligned} V(a, b, c) &= M(a, b) \to U(c, M(a, b)), \\ P(a, b, c) &= c \to M(c, b), \\ S_*(a, b, c) &= U(b, (M(a, b) \to P(a, b, c)) \wedge (M(c, b) \to c)), \\ Q_*(a, b, c) &= U(c, S_*(a, b, c) \to \neg P(a, b, c)), \\ G(a, b, c) &= \neg(V(a, b, c) \to Q_*(a, b, c)), \qquad B \wedge C = \neg(B \to \neg C). \end{aligned}$$

For an incoming store $\langle t, w, d, h\rangle$, define $A = c$, $D = d + x$, and $H = h + y$. The new command is exactly

$$\begin{aligned} \mathsf{AX}^{D}_{c,x,y}\langle t, w, d, h\rangle &= \langle K(H, D, A), W', 0, 0\rangle, \\ W' &= [\Gamma(D), h_6, h_5, h_4, h_3, h_2, h_1, \\ &\qquad \rho_8(D, A, H, d), U(d, h), t; w]. \end{aligned} \tag{D-AX}$$

Every head uses the displayed D, A, H; only the argument of the separate Γ has changed. Thus the command still pushes ten entries, counting the saved top, and still clears d, h. A failed gate guard does not remove the separately pushed $\Gamma(D)$.

This change is visible even on a zero incoming store. For the command parameters $(c, x, y) = (0, 1, 0)$, the former first stack entry is $\Gamma(0) = 20$, whereas the new one is $\Gamma(1) = 832$. The other emitted entries retain their former values. In particular, this is a change of the correlation between axiom parameters, not a global exchange of physical register names.

Let $\mathcal{C}_s$ be the supplied separate-head calculus: the six families above, dominated ρ_8, every $\Gamma(B)$, each guarded $G(a, b, c)$, modus ponens, and theorem generalization $B \mapsto U(i, B)$. "Theorem" means that there are no undischarged hypotheses; the formula may have free variable labels. The inherited calibration gives primitive recursive proof translations, with correctness provable in PA, establishing

$$\mathrm{PA} \vdash (\exists q\ \mathrm{Prf}_{\mathcal{C}_s}(q, 1) \ \leftrightarrow\ \neg\,\mathrm{Con}(\mathrm{PA}))\,. \tag{D-Cal}$$

This is the separate-head calibration retained in `references/logical_calibration/separate_head_calculus.tex`, together with its diagonal-calculus and PA195 dependencies [2]. PA194D uses precisely this calculus. It requires neither a new logical axiom nor a new translation of PA proofs.

7.2 All-store command identities and the changed selector

For this subsection, use the ordinary cached-C source with the new emitter and with its two original unconditional jumps. Its numerical commands have arbitrary stores $\langle t, w, d, h\rangle$; scratch registers obey the inherited clean command-boundary convention. Write S for cached Shift:

$$S\langle t, w, d, h\rangle = \begin{cases} \mathsf{halt}, & h = 1, \\ \langle h, \mathrm{tl}(w), d, \mathrm{hd}(w)\rangle, & h \neq 1. \end{cases}$$

For $y = d + x$, cached modus ponens is

$$\mathsf{MP}_x\langle t, w, d, h\rangle = \begin{cases} \langle 0, w, 0, y\rangle, & I(t, y) = h, \\ \langle 0, w, y, 0\rangle, & I(t, y) \neq h. \end{cases}$$

These are exactly the commands proved for the supplied cached controller [3].

Number the generated stack entries in (D-AX) by $q_1, \ldots, q_9$, so $q_1 = \Gamma(D)$, $q_8 = \rho_8(D, A, H, d)$, and $q_9 = U(d, h)$. The first eight entries and the gate have residue zero modulo four; q_9 has residue two. None is code one. Consequently the ten-entry stack calculations in the supplied cached command audit apply without any change in their exponents. Explicitly, define

$$\begin{aligned} \mathsf{Push} &= \mathsf{AX}^D_{0,0,0}; S^9; \mathsf{MP}_0, \\ \mathsf{Pop} &= \mathsf{AX}^D_{0,0,0}; S^{11}, \\ \mathsf{Discard} &= \mathsf{MP}_1; \mathsf{AX}^D_{1,0,0}; S^{10}; \mathsf{MP}_0. \end{aligned}$$

Their exact contracts are

$$\begin{aligned} \mathsf{Push}\langle t, w, d, h\rangle &= \langle 0, [t; w], 0, 0\rangle, \\ \mathsf{Pop}\langle t, [a; w], d, h\rangle &= \begin{cases} \mathsf{halt}, & t = 1, \\ \langle t, w, 0, a\rangle, & t \neq 1, \end{cases} \\ \mathsf{Discard}\langle t, w, 0, 0\rangle &= \langle 0, w, 0, 0\rangle. \end{aligned}$$

For example, $\mathsf{AX}^D; S^9$ leaves $\langle q_8, [t; w], 0, q_9\rangle$, and $\mathsf{AX}^D; S^{10}$ leaves $\langle q_9, w, 0, t\rangle$. The first ten Shifts test only zero and generated entries; the eleventh tests the saved top. Push's last MP fails because an implication has residue zero and q_9 has residue two. Discard's first MP fails since $I(t, 1) > 0$; after its AX and Shifts, the last MP fails since $I(U(1, 0), 0) > 0$. These proofs use the total projection equations $\mathrm{hd}[a; w] = a$ and $\mathrm{tl}[a; w] = w$, including the alias $[0; 0] = 0$. They do not presume that a zero push increases a stored stack length.

For $1 \leq j \leq 7$, define a selector whose argument order is D, H, A:

$$\mathsf{select}_j(D, H, A) = \mathsf{Discard}; \mathsf{AX}^D_{A,D,H}; S^{j+1}; \mathsf{Pop}^{9-j}.$$

It sends every $\langle t, w, 0, 0\rangle$ to $\langle q_j, w, 0, 0\rangle$. The desired change to the proof compiler is therefore exactly

$$\boxed{\Gamma(B): \qquad \mathsf{select}_1(B, 0, 0) = \mathsf{Discard}; \mathsf{AX}^D_{0,B,0}; S^2; \mathsf{Pop}^8.} \tag{D-Sel}$$

After Discard, $d = h = 0$, so this AX has $D = B$. The former selector used $\mathsf{select}_1(0, 0, B)$. Selection is safe even when $B = 0$ and even when the unrelated gate guard fails. The six retained heads use $j = 2, \ldots, 7$, with their original parameters and words.

For completeness, let Offset_0 be empty and let $\mathsf{Offset}_u = \mathsf{MP}_u$ for $u > 0$; from $\langle 0, w, 0, 0\rangle$ it produces $\langle 0, w, u, 0\rangle$. For $u \leq a$, the word

$$\mathsf{Discard}; \mathsf{Offset}_u; \mathsf{AX}^D_{b,a-u,c}; S^9; \mathsf{Pop}$$

selects $\rho_8(a, b, c, u)$. The word $\mathsf{Discard}; \mathsf{AX}^D_{c,b,a}; \mathsf{Pop}^{10}$ selects $K(a, b, c)$. Finally

$$\mathsf{Gen}_i = \mathsf{Push}; \mathsf{Offset}_i; S; \mathsf{AX}^D_{i,0,0}; S^{10}$$

sends $\langle a, w, 0, 0\rangle$ to $\langle U(i, a), w, 0, 0\rangle$, also when $a = 1$. The Shift before AX tests zero and puts a in the cache; AX clears that cache before a later Shift can test it.

7.3 A strong proof compiler and both logical directions

The cached-MP suffix is

$$\mathsf{MPSuffix}_b = S; S; \mathsf{MP}_b; S.$$

From $\langle 0, [a, I(a,b), 0; w], 0, 0\rangle$, it halts if $a = 1$ or $b = 1$, and otherwise returns $\langle b, w, 0, 0\rangle$. The second Shift tests a; successful MP places b in the cache; the final Shift tests b and retrieves the dummy zero. The dummy is necessary to preserve the underlying w. This calculation also holds at the numerical zero-stack alias.

Compile axiom leaves with the selectors just given, and compile a generalization node by appending Gen_i to its premise word. If an MP node has minor proof p_0 of a and major proof p_1 of $I(a,b)$, use

$$W_p = \mathsf{Discard}; \mathsf{Push}; W_{p_1}; \mathsf{Push}; W_{p_0}; \mathsf{Push}; \mathsf{MPSuffix}_b.$$

Lemma 7.1 (Uniform Gamma-D proof compiler)**.** *For every finite $\mathcal{C}_s$-proof p and every t, w, the word W_p from $\langle t, w, 0, 0\rangle$ either halts or returns $\langle \mathrm{concl}(p), w, 0, 0\rangle$. If p proves code one, then*

$$\mathsf{AX}^{D}_{0,0,0}; W_p; \mathsf{Pop}$$

halts from every incoming four-component store.

Proof. Induct on the proof tree. The selectors prove the axiom case, including the changed selector (D-Sel). The generalization contract proves that case. In the MP case, Discard and Push introduce a dummy zero; if neither recursive premise word halts, their contracts and Push produce precisely the suffix's three-entry precondition. The suffix gives either an early halt or the required conclusion with the original tail. An earlier halt remains a halt when a word is extended. For the second claim, the initial AX clears both caches, so the first claim applies to its arbitrary resulting top and stack. If no earlier halt occurs, the last Pop tests the derived top one before inspecting the arbitrary underlying stack. □

This all-store clause matters because the inherited exhaustive command-word scheduler retains its stores between trials. A refuting proof yields a finite universally halting word, so the scheduler either halts earlier or halts when it reaches this word. The inherited word encoding and scheduler proof are unchanged.

Conversely, annotate t, h, and each decoded stack entry with a $\mathcal{C}_s$-derivation. Code zero has the fixed proof $\rho_8(0,0,0,0) = 0$, which supplies all zero padding. AX preserves these annotations: the six heads and a successful gate are axioms; a failed gate is zero; $\Gamma(D)$ is an unconditional instance of the existing diagonal family; $d \le d + x$ permits the propositional head; and $U(d,h)$ follows by generalization from the old cache. A successful MP derives $d + x$ from the annotations for t and $h = I(t, d + x)$; a failed MP returns zero. Shift moves derived entries, and a halting Shift exposes a proof of one. No derivability condition is required of the offset d. The actual seeded source begins with the required zero annotations. This proves that its halting yields a refutation.

All selectors and proof-word constructors are primitive recursive. The arguments are inductions on proof trees, finite lists, source loop counters, and traces. Together with the unchanged scheduler they prove in PA that the ordinary Gamma-D cached-C source halts exactly when $\mathcal{C}_s$ has a refutation. The proof translations in (D-Cal) then give its equivalence to $\neg\,\mathrm{Con}(\mathrm{PA})$. In particular, the soundness direction does not appeal to the standard truth of arithmetic or to reflection for PA.

7.4 Exact source binding and the two zero tests

Let P_D be that ordinary cached-C Gamma-D source and let P_D^Z be the source implemented by PA194D. Both have 3,649 primitive sites, numbered from zero. Relative to the old 190-token Gamma-A emitter, the copy macros at token indices 121, 122, and 123 change from `copyA` to `copyD`. Their source spans are respectively 2258–2283, 2284–2309, and 2310–2335. Each span has 26 primitive instructions. The six altered register arguments, and all their exact successors, are

Site	Old instruction	Gamma-D instruction
2277	$\mathrm{dec}(3, 2281, 2278)$	$\mathrm{dec}(4, 2281, 2278)$
2282	$\mathrm{inc}(3, 2283)$	$\mathrm{inc}(4, 2283)$
2303	$\mathrm{dec}(3, 2307, 2304)$	$\mathrm{dec}(4, 2307, 2304)$
2308	$\mathrm{inc}(3, 2309)$	$\mathrm{inc}(4, 2309)$
2329	$\mathrm{dec}(3, 2333, 2330)$	$\mathrm{dec}(4, 2333, 2330)$
2334	$\mathrm{inc}(3, 2335)$	$\mathrm{inc}(4, 2335)$

Here registers 3 and 4 are logical A and D, respectively. Every other instruction of P_D, including every successor, agrees with the old cached-C source. A separate free-expression calculation checks the entire stack and gate of (D-AX) for symbolic incoming parameters.

The physical search reassociates the emitter tree. The winning grammar's complete resolved instruction graph equals that of an independently balanced tree over the same macro word. It also equals the graph obtained by changing only these macros in the supplied accepted association. Thus the binding checks structured jump destinations, not only equality of the flattened token sequence.

To pass from P_D to P_D^Z, change precisely

site	P_D	P_D^Z
26	$\mathrm{jump}(34)$	$\mathrm{dec}(6, 34, 27)$
3553	$\mathrm{jump}(3637)$	$\mathrm{dec}(4, 3637, 3554)$.

The first tests scratch X; the second is the final AX return and tests D. This adds two branch outcomes: P_D has 4,749, and P_D^Z has 4,751.

Use the positivity abstraction of Section 4, now computed for P_D from the same clean seed

$$c_* = (25, (2, 0, 0, 0, 0, 0, 0, 0, 0)).$$

Its least abstract closure R_D contains 263,244 location/mask pairs. A separate replay verifies that the seed belongs to it, that it is closed, that the two source programs have exactly the same abstract successors on it, and that the tested bits are zero at the two changed sites. There are 32 masks at site 26, all with bit 6 clear, and eight masks at site 3553, all with bit 4 clear. The sorted-pair certificate has SHA-256

`57c58dcae511fe13a6c200e7cfe4f4322370aec72b4a2fe5ba39b906fe62dbf8`.

This is a fresh invariant for the changed formula emitter; it is not an assumption that the Gamma-A invariant still applies.

The abstraction lemma and induction on execution length therefore prove exact equality of the P_D and P_D^Z source runs from c_*, until and including any halt. Each inserted decrement sees zero, preserves every register, and takes the former jump target. PA proves the finite closure facts and the two inductions, hence

$$\mathrm{PA} \vdash (\mathrm{Halt}(P_D, c_*) \leftrightarrow \mathrm{Halt}(P_D^Z, c_*)).$$

This source equivalence is claimed from the specified seed. The arbitrary-store command calculations above concern P_D; they are not being inferred for arbitrary inputs to the specialized source P_D^Z.

7.5 Literal realization and the PA theorem

The nine folds are performed simultaneously on the 203-control raw allocation. In native labels their replacement rows are:

Removed	Retained	New read-0 action	New read-1 action
2004	2063	$0\,L\,2049$	$1\,R\,90$
2015	2023	$0\,L\,2061$	$1\,R\,2012$
2018	2012	$0\,L\,2055$	$1\,R\,90$
2022	2033	$0\,L\,2002$	$1\,R\,61$
2025	2046	$0\,L\,2034$	$1\,R\,2029$
2028	2039	$0\,L\,2014$	$1\,L\,2003$
2037	2059	$0\,L\,2044$	$1\,R\,2038$
2053	2011	$0\,R\,98$	$1\,L\,2019$
2064	2032	$0\,L\,2017$	$1\,L\,2041$

Every incoming row establishes the required predecessor bit. None of the folded controls is the initial control. The 36 local cases are checked on the final simultaneous quotient: 14 take one step, 19 take three, and three take five. The traces preserve the exterior tape and the full required local effect; the supplied universal source contracts also verify the resulting whole interpreter independently.

The literal table has an independently checked physical simulation contract for each of the 4,751 primitive outcomes of P_D^Z, including the positive branches that the seeded source invariant excludes. These contracts range over arbitrary natural primary values and header gap and every finite nonempty spare list. The independent checker uses 1,091,154 symbolic steps, with a maximum of 385 for one case; every duration has nonnegative coefficients and constant at least three. It derives four list lemmas from the literal rows and invokes no arithmetic-box oracle. The nine compatible reader folds have 36 local all-tape cases. The exact blank initialization takes 58,168 literal transitions to the complete frame for c_*, in native control 84 with head at -2. The frame has source path 1000000, header distance 48, primary vector $(2,0,0,0,0,0,0,0,0)$, and 26 spare blocks of value one.

The source simulations terminate in positive time. Consequently a source halt produces a physical halt, and a physical halt cannot precede a nonhalting concatenation of more positive source simulations than the alleged halting time. As in Section 5, this argument requires no inverse decoding of arbitrary intermediate tapes. The finite startup and local certificates, and the unary, list and trace inductions, are formalizable in PA.

Theorem 7.2 (Gamma-D consistency machine)**.** *Let H_D be the arithmetic sentence asserting a finite halting trace of the specified PA194D table from its blank start. Then*

$$\mathrm{PA} \vdash \big(H_D \leftrightarrow \neg\,\mathrm{Con}(\mathrm{PA})\big), \qquad \mathrm{PA} \vdash \big(\neg H_D \leftrightarrow \mathrm{Con}(\mathrm{PA})\big).$$

Proof. Compose, inside PA, the physical simulation, the seeded zero-test equivalence, the soundness and strong proof compiler, and the supplied proof calibration:

$$\begin{aligned} H_D &\longleftrightarrow \mathrm{Halt}(P_D^Z, c_*) \longleftrightarrow \mathrm{Halt}(P_D, c_*) \\ &\longleftrightarrow \exists q\ \mathrm{Prf}_{\mathcal{C}_s}(q,1) \longleftrightarrow \neg\,\mathrm{Con}(\mathrm{PA}). \end{aligned}$$

Classical logic in PA gives the nonhalting formulation. □

The theorem is an ordinary proof with explicit finite and symbolic certificates, using the retained ordinary PA calibration. It is not a new proof-assistant kernel export or an iPRA equivalence theorem. It makes no consistency assumption in establishing the PA-provable biconditional. PA194D supplies a second construction at the requested 194-state fallback threshold; it does not lower that threshold.

8 Novelty within the pinned corpus

For a deterministic binary action graph with a chosen root, breadth-first discovery in read-bit order produces a canonical numbering. To forget the root, the checker takes the lexicographically least serialization over all controls from which the entire working graph is reachable. The unique terminal control is marked separately. Two such graphs are isomorphic exactly when these canonical serializations agree.

For the 194-state comparison, the calculation is repeated after global L/R reflection, global interchange of tape symbols, and their combination. Even allowing a change of designated start and the symbol-exchanged blank convention, PA194J matches none of the eight existing 194-state classes. Graphs of other cardinalities cannot be isomorphic to it. The four variants choosing A, D, H or X at the final zero-test location are also mutually distinct under the same tests; the delivered principal table uses D.

The finite comparison establishes novelty relative to the retrieved corpus. It is not a claim of worldwide priority. Its semantic significance comes from the independently proved source changes and actual startup use of changed rows, not from a hash inequality alone.

The independently audited PA194D graph and its CSZ sibling also differ from all eight old graphs, from each other, and from all four new J-family graphs under these same transformations. A separate free-term audit of the entire emitter rules out every global permutation of A, D, H: exactly one of its ten pushed entries changes, while the other nine, parameter reads and gate agree. The concrete head values 20 and 832 in Section 7 further expose that logical difference.

9 Searches below 194 and their limits

The following new searches jointly mutate source associations and evaluate compatible finite-history folds. Closing-leaf searches additionally vary the final jump between ordinary return and candidate zero-register decrements. The score is the effective reader count plus scanner/address penalties; it is a filter, not a physical correctness certificate. A score below 60 would be a candidate for fewer than 194 states with the 134-control fixed part. Every such candidate would still require the complete source and tape proofs.

Search	Mutations	Best score	Seconds
CSZ	11,263,380	60	900
CZ	9,991,239	60	900
CSZ103A	2,940,290	60	450
CSZ103X	3,442,575	60	450
closing103A	4,489,001	60	450
negation_search	3,062,902	60	450
h1D	890,683	67	180
h1D_adaptive	1,128,819	64	240
separateD	3,608,903	62	450
separateD_adaptive	1,927,245	60	240
separateH	3,447,089	63	450
separateH_adaptive	4,398,267	63	240
warm_D_CSZ_A	9,288,577	60	480
warm_D_CSZ_X	9,325,004	60	480

Logs, frozen input pools, random seeds and C++ source are retained under `research/search/`. Wall-time stopping means mutation counts vary with the host. No run produced a score below 60. The adaptive separate-D run reached 69 readers and nine simultaneous folds; this candidate became PA194D only after the independent proofs reported in Section 7. Failed searches establish no lower bound and do not exhaust all machines, logical calculi, controller layouts, or source associations.

The new work also explored changes to the logical emitter. Three 191-token nine-entry variants relocate a diagonal antecedent into a quantifier head; two 190-token ten-entry variants emit $\Gamma(D)$ or $\Gamma(H)$ instead of $\Gamma(A)$. Exact free-algebra word checks, primitive source graphs, proof arguments and discovery seeds are under `research/logical_variants/`. The $\Gamma(D)$ search reached the second 194-state construction given above. The other four variants remain logical source proposals and search inputs, without a claimed 194-state physical realization. PA194J uses the unchanged parent numerical calculus; PA194D supplies the new selector calibration.

9.1 Two concrete failures worth preserving

Deleting scanner control 34 and redirecting it to 75 fails 113 of the 4,780 universal contracts for the supplied primary PA194 source. The failures include fixed controller paths, so shortening AX addresses alone would not justify the deletion. A finite blank trace that has not yet encountered a bad address is insufficient evidence.

The ZFC zero-spare idea is also not a plug-in change. Altering native row 198/0 from $1L40$ to $0L40$ in PA194CS fails its bootstrap: it visits the source dispatcher only at times 38 and 66 before entering an unbounded rightward one-writing drift. The ZFC construction had a different scheduler/bootstrap proof and cannot supply the missing PA initialization. This experiment is excluded from the delivered table.

9.2 What transfers from ZFC and concise iPRA

Useful ZFC methods [6] already present in the PA frontier include contextual reader folds, source reassociation with exact control-flow binding, cached conclusions, and exploiting destructive comparison symmetry. The new construction extends the global zero-information method beyond the single scheduler jump found in PA194CZ [4]. Zero-spare and command-credit ideas remain possible research directions only after a new bootstrap and command proof are supplied.

ConciseIPRA [7] offers small proof presentations, primitive-recursive syntax, and disciplined separation of logical and executable certificates. Those ideas help organize a verifier, but replacing the checked theory by iPRA would change the consistency statement. A smaller iPRA checker does not directly give a smaller machine for Con(PA). The present construction therefore preserves the PA-calibrated calculus.

10 Reproduction and evidence boundaries

Python 3.10 or later and its standard library suffice for the accepted construction and checks. From the archive root, run

```
python3 -B verify.py
```

Assertions must be enabled; the scripts reject optimization mode. The full check rebuilds both literal tables and sources, recomputes the finite invariants, runs both independent physical proofs, replays the whole blank initializations, and checks corpus novelty and the changed logical emitter. It works in temporary copies to preserve the frozen packet. The optional `--quick` mode omits the universal physical replays; it retains the other checks. Optional C++ searches are not required. The saved receipts give independently reviewable results but do not replace execution of the verifier.

To rebuild this article, run `build_report.sh`; it invokes `pdflatex` twice with failure-on-error flags. The exact table, source files, supporting proof texts and finite receipts are included. `SHA256SUMS` binds the delivery. The attribution and license files identify the inherited James Chang, ProveIt and symbolic-engine material.

The claims should be read at their actual levels:

- Fresh finite facts: state count, table hashes, source maps, graph distinctions, mask closure, and complete startup equality.
- Fresh symbolic mathematics: arbitrary-register physical contracts, four list lemmas and positive simulation durations.
- New ordinary theorem: exact seeded source preservation and the resulting PA-provable halting equivalence with the parent.
- Inherited ordinary theorem: the numerical calculus, strong command compiler and scheduler are calibrated to PA consistency.
- Unclaimed: a smaller machine, a general lower bound, a new end-to-end proof-assistant theorem, or an iPRA proof of the full consistency equivalence.

11 Further research questions

1. Can the new helper pairs $(39, 21)$, $(108, 17)$, and $(221, 144)$ support a reader allocation of effective size 59? With the current 134-control fixed part, that would give 193 states. A search must include the source invariant and all physical branch contracts, not just the new pair identities.
2. Can one combine two different known-zero registers in a larger decoder gadget to replace three readers by two? The four certified final-register variants provide different interfaces for such a search. Longer detours would need a positive local simulation proof.

3. Can the fixed controller be reassociated to shorten *all* return addresses, including startup and the equality guard, enough to remove one scanner control? AX-only address constraints miss the current fixed-path obstruction.

4. Can a smaller finite invariant or a structured loop invariant replace the 263,180-pair certificate? The current set is easy to recompute but is not optimized as a mathematical or Lean artifact.

5. Can the successful $\Gamma(D)$ allocation of 69 readers and nine compatible folds be combined with the new PA194J helper pairs to obtain a tenth fold? Can the other four logical emitter variants gain comparable sharing? Their equiconsistency arguments alone do not determine the physical-state cost.

6. Can the new invariant theorem and the literal tape contracts be connected to the repository's existing Lean source and arithmetic endpoints? A first useful target is the seeded source equivalence, whose proof needs only finite closure and induction; the full iPRA calibration remains a separate task.

References

[1] Vladimir Reshetnikov, *ProveIt*, pinned revision `17f92105e2bc4317feae6c219cb2143437b482ff`. https://github.com/VladimirReshetnikov/ProveIt/tree/17f92105e2bc4317feae6c219cb2143437b482ff/Computability/BusyBeaver/Research/PA.

[2] Supplied PA194 research packets, 13 November 2026, filed under `Computability/BusyBeaver/Research/PA/docs/pa194_conpa_2026-11-13` and its `_2` companion. Exact source and table identities are in the supplied comparison ledger.

[3] Supplied *Two new 194-state PA consistency machines*, `docs/incoming/pa194_new_designs_2026-11-13.zip`, especially `secondary/pa194CS` and `construction/logical_reference`.

[4] Supplied PA194C/PA194CZ packet, `docs/incoming/pa194c_conpa_2026-11-13(1).zip`, especially `construction/build194cz.py`.

[5] James Chang, PA196 and PA209 developments, imported snapshots under `PA/jamesyc-upstream`; original attribution and licenses are preserved in the supplied sources.

[6] ProveIt ZFC consistency-machine research, especially `candidate-k215/alternatives/zero-credit` and its related source and physical preservation arguments.

[7] ProveIt, *Concise presentations of intuitionistic primitive recursive arithmetic*, https://github.com/VladimirReshetnikov/ProveIt/tree/17f92105e2bc4317feae6c219cb2143437b482ff/Logic/PrimitiveRecursiveArithmetic/ConciseIPRA.

A Complete dense transition table

Each entry $w\,D\,q'$ means: write w, move in direction D, and enter q'. The initial state is 0 and terminal halt label is 194. The paired columns contain all 194 working states, with both read symbols.

State	Read 0	Read 1	State	Read 0	Read 1
0	1 R 1	1 R 2	97	0 L 112	1 R 113
1	1 R 3	1 R 1	98	0 R 114	0 L 14
2	0 R 0	1 R 2	99	0 R 115	1 R 91
3	0 R 4	0 R 1	100	0 R 115	1 R 106
4	1 R 5	1 L 6	101	0 R 116	0 L 14
5	1 L 7	1 L 8	102	0 R 68	1 L 66
6	0 L 7	1 L 6	103	0 R 117	1 R 118
7	0 L 9	1 L 6	104	0 R 89	1 R 104
8	0 L 10	0 L 11	105	0 R 14	1 R 105
9	1 R 12	0 L 9	106	0 L 112	1 L 33
10	0 L 9	0 L 10	107	0 R 67	1 R 119
11	1 L 8	1 L 11	108	0 R 63	1 R 108
12	0 R 12	0 R 13	109	0 R 120	1 L 121
13	0 L 14	0 R 13	110	0 R 122	1 R 123
14	1 L 15	0 L 15	111	0 R 46	0 L 14
15	0 L 15	1 L 16	112	0 R 86	1 R 96
16	0 L 17	1 L 16	113	0 R 112	1 L 91
17	0 L 18	1 L 16	114	0 R 124	1 R 124
18	0 L 19	1 L 16	115	0 L 60	0 R 83
19	0 L 20	1 L 16	116	0 R 62	0 R 89
20	0 L 21	1 L 16	117	0 L 81	1 R 125
21	0 L 22	1 L 16	118	0 R 126	1 R 127
22	0 L 23	1 L 16	119	0 R 47	1 R 85
23	0 R 23	1 R 24	120	0 R 55	0 R 108
24	0 R 25	1 R 26	121	0 R 35	1 R 33
25	0 R 27	1 R 28	122	0 R 95	1 R 111
26	0 R 29	1 R 30	123	0 R 75	1 R 74
27	0 R 31	1 R 194	124	0 R 46	1 R 46
28	0 L 32	1 R 33	125	0 R 128	1 R 129
29	0 R 34	1 R 35	126	0 R 32	1 R 64
30	0 L 33	1 L 36	127	0 R 31	1 R 109
31	0 R 37	1 R 38	128	0 R 130	1 R 131
32	0 R 39	1 R 40	129	0 R 132	1 R 133
33	0 R 41	1 R 42	130	0 R 134	1 R 135
34	0 R 43	1 R 44	131	0 R 136	1 R 137
35	0 R 45	1 R 36	132	0 R 138	1 R 139
36	0 R 46	1 R 47	133	0 R 140	1 R 141
37	0 R 48	1 R 41	134	0 R 93	1 R 121
38	0 R 49	1 R 50	135	0 R 142	1 R 124
39	0 R 51	1 R 52	136	0 R 143	1 R 144
40	0 R 53	1 R 54	137	0 R 145	1 R 146
41	0 R 55	0 R 56	138	0 R 134	1 R 147
42	0 R 57	0 L 14	139	0 R 46	1 R 148
43	0 R 58	1 R 59	140	0 R 149	1 L 131
44	0 R 60	1 R 49	141	0 R 150	1 R 151
45	0 R 55	0 R 61	142	0 R 72	1 R 40
46	0 R 62	0 R 63	143	0 R 28	1 R 152

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State	Read 0	Read 1	State	Read 0	Read 1
47	0 *L* 64	1 *L* 65	144	0 *L* 153	1 *L* 154
48	0 *R* 45	1 *R* 50	145	0 *L* 129	1 *R* 155
49	0 *L* 66	0 *L* 15	146	0 *R* 156	1 *R* 157
50	0 *R* 67	1 *R* 68	147	0 *R* 149	1 *R* 158
51	0 *R* 69	1 *R* 70	148	0 *R* 159	1 *R* 46
52	0 *R* 71	1 *R* 72	149	0 *R* 93	1 *R* 160
53	0 *R* 73	1 *R* 74	150	0 *R* 161	1 *R* 162
54	0 *R* 75	1 *R* 76	151	0 *R* 163	1 *R* 120
55	0 *L* 77	0 *R* 1	152	0 *R* 164	1 *R* 165
56	0 *R* 61	1 *R* 56	153	0 *R* 166	1 *R* 157
57	0 *L* 69	1 *L* 35	154	0 *R* 167	1 *R* 156
58	0 *R* 78	1 *R* 79	155	0 *R* 135	1 *R* 130
59	0 *R* 51	1 *R* 80	156	0 *R* 147	1 *R* 40
60	0 *L* 81	1 *R* 82	157	0 *R* 134	1 *R* 168
61	0 *R* 83	1 *R* 61	158	0 *R* 40	1 *L* 169
62	0 *L* 77	0 *R* 2	159	0 *L* 170	1 *R* 171
63	0 *R* 84	1 *R* 63	160	0 *R* 102	1 *R* 80
64	0 *R* 30	1 *R* 85	161	0 *R* 172	1 *R* 170
65	0 *R* 86	1 *R* 87	162	0 *R* 173	1 *R* 174
66	0 *R* 35	1 *R* 88	163	0 *R* 175	1 *R* 176
67	0 *R* 41	0 *L* 13	164	0 *R* 177	1 *R* 178
68	0 *R* 55	0 *R* 89	165	0 *L* 179	1 *L* 180
69	0 *R* 90	1 *R* 65	166	0 *R* 149	1 *R* 140
70	0 *R* 47	1 *R* 91	167	0 *R* 165	1 *R* 181
71	0 *R* 41	1 *R* 92	168	0 *L* 130	1 *R* 46
72	0 *R* 93	1 *R* 94	169	0 *R* 182	1 *R* 135
73	0 *R* 95	1 *L* 96	170	0 *R* 183	1 *R* 145
74	0 *R* 47	1 *R* 73	171	0 *R* 158	1 *R* 130
75	0 *R* 88	1 *R* 97	172	0 *R* 139	1 *R* 165
76	0 *R* 47	1 *R* 98	173	0 *R* 158	1 *R* 184
77	1 *R* 77	1 *L* 15	174	0 *R* 185	1 *R* 102
78	0 *R* 90	1 *R* 41	175	0 *R* 31	1 *R* 49
79	0 *R* 99	1 *R* 100	176	0 *R* 186	1 *R* 175
80	0 *R* 41	1 *R* 101	177	0 *R* 134	1 *R* 142
81	0 *R* 102	1 *R* 85	178	0 *R* 32	1 *R* 187
82	0 *R* 103	0 *L* 67	179	0 *R* 144	1 *R* 188
83	0 *R* 104	1 *R* 83	180	0 *R* 189	1 *R* 190
84	0 *R* 105	1 *R* 84	181	0 *R* 40	1 *R* 191
85	0 *R* 106	0 *L* 14	182	0 *R* 168	1 *R* 40
86	0 *R* 62	0 *R* 104	183	0 *R* 192	1 *R* 46
87	0 *R* 107	0 *L* 85	184	0 *R* 153	1 *R* 146
88	0 *R* 30	1 *R* 57	185	0 *R* 165	1 *R* 180
89	0 *R* 108	1 *R* 89	186	0 *R* 119	1 *R* 109
90	0 *L* 60	0 *R* 84	187	0 *R* 193	1 *R* 159
91	0 *R* 90	1 *L* 109	188	0 *R* 169	1 *R* 154
92	0 *R* 97	0 *L* 14	189	0 *R* 158	1 *R* 192
93	0 *R* 64	1 *R* 110	190	0 *R* 179	1 *R* 182
94	0 *R* 109	1 *R* 71	191	0 *R* 182	1 *R* 40
95	0 *L* 60	0 *R* 105	192	0 *R* 149	1 *R* 168
96	0 *L* 59	1 *R* 111	193	0 *R* 72	1 *R* 144

B Complete dense PA194D transition table

The second table uses the same entry notation and machine convention as Appendix A. Its initial state is 0 and terminal label is 194. All 388 rows agree with `logical_emitter/machine/pa194d.tm`.

State	Read 0	Read 1	State	Read 0	Read 1
0	1 *R* 1	1 *R* 2	97	0 *R* 36	1 *R* 112
1	1 *R* 3	1 *R* 1	98	0 *R* 111	1 *R* 75
2	0 *R* 0	1 *R* 2	99	0 *L* 63	0 *R* 109
3	0 *R* 4	0 *R* 1	100	0 *L* 62	1 *R* 113
4	1 *R* 5	1 *L* 6	101	0 *L* 90	1 *R* 114
5	1 *L* 7	1 *L* 8	102	0 *R* 115	0 *L* 14
6	0 *L* 7	1 *L* 6	103	0 *R* 69	1 *R* 103
7	0 *L* 9	1 *L* 6	104	0 *R* 116	1 *R* 95
8	0 *L* 10	0 *L* 11	105	0 *R* 116	1 *R* 67
9	1 *R* 12	0 *L* 9	106	0 *R* 117	0 *L* 14
10	0 *L* 9	0 *L* 10	107	0 *R* 118	1 *L* 71
11	1 *L* 8	1 *L* 11	108	0 *R* 119	1 *R* 120
12	0 *R* 12	0 *R* 13	109	0 *R* 14	1 *R* 109
13	0 *L* 14	0 *R* 13	110	0 *R* 66	1 *R* 110
14	1 *L* 15	0 *L* 15	111	0 *R* 121	1 *L* 35
15	0 *L* 15	1 *L* 16	112	0 *R* 122	1 *R* 123
16	0 *L* 17	1 *L* 16	113	0 *R* 88	0 *L* 14
17	0 *L* 18	1 *L* 16	114	0 *R* 90	1 *L* 95
18	0 *L* 19	1 *L* 16	115	0 *R* 124	1 *R* 124
19	0 *L* 20	1 *L* 16	116	0 *L* 63	0 *R* 103
20	0 *L* 21	1 *L* 16	117	0 *R* 68	0 *R* 91
21	0 *L* 22	1 *L* 16	118	0 *R* 58	0 *R* 91
22	0 *L* 23	1 *L* 16	119	0 *L* 86	1 *R* 125
23	0 *R* 23	1 *R* 24	120	0 *R* 126	1 *R* 127
24	0 *R* 25	1 *R* 26	121	0 *R* 58	0 *R* 110
25	0 *R* 27	1 *R* 28	122	0 *R* 99	1 *R* 113
26	0 *R* 29	1 *R* 30	123	0 *R* 79	1 *R* 78
27	0 *R* 31	1 *R* 194	124	0 *R* 88	1 *R* 88
28	0 *L* 32	1 *R* 33	125	0 *R* 128	1 *R* 129
29	0 *R* 34	1 *R* 35	126	0 *R* 32	1 *R* 36
30	0 *L* 36	1 *L* 37	127	0 *R* 31	1 *R* 111
31	0 *R* 38	1 *R* 39	128	0 *R* 130	1 *R* 131
32	0 *R* 40	1 *R* 41	129	0 *R* 132	1 *R* 133
33	0 *R* 42	1 *R* 43	130	0 *R* 134	1 *R* 135
34	0 *R* 44	1 *R* 45	131	0 *R* 136	1 *R* 137
35	0 *R* 46	1 *R* 33	132	0 *R* 88	1 *R* 138
36	0 *R* 47	1 *R* 48	133	0 *R* 139	1 *R* 140
37	0 *R* 49	1 *R* 50	134	0 *R* 97	1 *R* 35
38	0 *R* 51	1 *R* 42	135	0 *R* 141	1 *R* 124
39	0 *R* 52	1 *R* 53	136	0 *R* 142	1 *R* 143
40	0 *R* 54	1 *R* 55	137	0 *R* 144	1 *R* 145
41	0 *R* 56	1 *R* 57	138	0 *L* 137	1 *R* 146
42	0 *R* 58	0 *R* 59	139	0 *L* 147	1 *R* 41
43	0 *R* 60	0 *L* 14	140	0 *R* 148	1 *R* 149
44	0 *R* 61	1 *R* 62	141	0 *L* 148	1 *R* 41
45	0 *R* 63	1 *R* 52	142	0 *R* 28	1 *R* 150
46	0 *R* 64	1 *R* 65	143	0 *R* 151	1 *R* 152

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State	Read 0	Read 1	State	Read 0	Read 1
47	0 R 58	0 R 66	144	0 R 153	1 R 154
48	0 R 67	0 L 14	145	0 R 155	1 R 156
49	0 R 68	0 R 69	146	0 R 135	1 R 130
50	0 R 70	0 L 48	147	0 R 157	1 R 158
51	0 R 64	1 R 53	148	0 R 159	1 R 160
52	0 L 71	0 L 15	149	0 R 161	1 R 121
53	0 R 72	0 L 10	150	0 R 162	1 R 163
54	0 R 73	1 R 74	151	0 R 155	1 R 164
55	0 R 75	1 R 76	152	0 R 138	1 R 165
56	0 R 77	1 R 78	153	0 R 134	1 R 166
57	0 R 79	1 R 80	154	0 R 167	1 R 168
58	0 L 81	0 R 1	155	0 L 169	1 R 139
59	0 R 82	1 R 59	156	0 R 170	1 R 171
60	0 L 73	1 L 46	157	0 R 41	1 R 158
61	0 R 83	1 R 84	158	0 L 172	1 L 156
62	0 R 54	1 R 85	159	0 R 76	1 R 173
63	0 L 86	1 R 87	160	0 R 158	1 R 107
64	0 R 58	0 R 82	161	0 R 174	1 R 175
65	0 R 88	1 R 30	162	0 R 134	1 R 176
66	0 R 89	1 R 66	163	0 R 76	1 L 177
67	0 L 90	1 L 33	164	0 R 172	1 R 177
68	0 L 81	0 R 2	165	0 R 166	1 R 157
69	0 R 91	1 R 69	166	0 R 168	1 R 124
70	0 R 72	1 R 92	167	0 L 129	1 R 171
71	0 R 46	1 R 93	168	0 L 151	1 L 135
72	0 R 42	0 L 13	169	0 R 178	1 R 179
73	0 R 94	1 R 37	170	0 R 180	1 R 181
74	0 R 30	1 R 95	171	0 R 182	1 R 183
75	0 R 42	1 R 96	172	0 R 139	1 R 130
76	0 R 97	1 R 98	173	0 R 184	1 R 185
77	0 R 99	1 L 100	174	0 R 31	1 R 52
78	0 R 30	1 R 77	175	0 R 186	1 R 174
79	0 R 93	1 R 101	176	0 R 141	1 R 32
80	0 R 30	1 R 102	177	0 R 179	1 R 155
81	1 R 81	1 L 15	178	0 R 182	1 R 141
82	0 R 103	1 R 82	179	0 R 181	1 R 135
83	0 R 94	1 R 42	180	0 R 154	1 R 179
84	0 R 104	1 R 105	181	0 R 183	1 R 41
85	0 R 42	1 R 106	182	0 R 97	1 R 187
86	0 R 107	1 R 48	183	0 L 130	1 R 88
87	0 R 108	0 L 72	184	0 R 163	1 R 188
88	0 R 68	0 R 66	185	0 R 189	1 R 190
89	0 R 109	1 R 89	186	0 R 92	1 R 111
90	0 R 49	1 R 100	187	0 R 107	1 R 85
91	0 R 110	1 R 91	188	0 R 147	1 R 165
92	0 R 30	1 R 48	189	0 R 191	1 R 167
93	0 R 47	1 R 60	190	0 R 169	1 R 192
94	0 L 63	0 R 89	191	0 R 182	1 R 171
95	0 R 94	1 L 111	192	0 R 193	1 R 166
96	0 R 101	0 L 14	193	0 R 191	1 R 141

C All 41 research packets

The following ledger records the principal contribution of each packet. Historical semantic claims are attributed to their supplied reports; literal table identities and the present frontier comparison were freshly checked. Full locators, archive identities and detailed evidence statuses are in the CSV/JSON ledger.

Packet	Principal contribution or outcome
`Below_U208_Research_2026-10-29.zip`	Full-frame windows, offset quotients and guarded pruning; source 3,909→3,904; no accepted below-208 table.
`ConPA_below208_research_2026-10-29.zip`	Finite-model role-collapse objection, 279,936 fixed-association parameter cases, fold synthesis; no sub-208.
`PA195_ConPA_research_2026-11-14.zip`	Positive 195S and 195C: unchanged separate 190-token AX word, 67−6 readers; full source/runtime/startup certificates plus independent literal-tape audits.
`PA_consistency_TM_research_2026-11-13.zip`	159-node five-zero source and 157-node combined representation; exact source CFG checks, no smaller physical allocation.
`U208_ConPA_2026-10-29.zip`	Particular native controls 172/211 history fold reduces a U209 parent to U208.
`U208_ConPA_Conditional_Context_2026-10-29.zip`	Conditional-entry proof; same two original U208 graph classes.
`U208_ConPA_Entry_Context_2026-10-29.zip`	Full entry-context proof; same two original classes.
`U208_ConPA_History_2026-10-29.zip`	Finite-history proof with four exceptional blocks; same original saving.
`conpa_tm_comparison_and_reduction_search.zip`	Experimental MP-result-in-H and MP-without-zero-push 196 variants; rejected Shift layouts halt at 43,563.
`conpa_tm_comparison_and_search.zip`	A/D register-swap 196; rejected two-rotation 195 fails 93 contracts.
`conpa_tm_study_2026-11-13.zip`	Combined196 = H-only Shift + MP-result-in-H: 3,788 sites, 4,928 contracts, 57,922 startup; direct-negation 197 too.
`pa194C_conpa_2026-11-13(2).zip`	PA194C; same literal cached-head machine as other C packets. Broad comparison and below194 searches.
`pa194_conpa_2026-11-13(2).zip`	Different separate 190-token association: 68 readers and 8 simultaneous folds yield a third positive194 graph. Also contains a distinct intermediate195 separate 190-token layout with 67 readers and6 folds. The actual intermediate source is3673 sites/4780 contracts despite a clerical report-table diagonal/3700 label.
`pa194_conpa_2026-11-13.zip`	Three-way saving composition: separate 190-token emitter, 67 raw AX readers and 7 simultaneous folds yield primary PA194; the A/D register permutation gives a second nonisomorphic194. Both include complete 4780 symbolic contracts, exact startup, and a written PA calibration.
`pa194_new_designs_2026-11-13.zip`	PA194C plus PA194CS. CS uses equality symmetry to remove three source sites, adds a reader and folds41 into330, remaining194.
`pa194c_conpa.zip`	PA194C; also eight-fold cached variant pa194altC as checked search input, not principal theorem.
`pa194c_conpa_2026-11-13(1).zip`	PA194C plus PA194CZ zero-register-jump variant at source 26, justified by263244 positivity-mask pairs.
`pa194c_conpa_2026-11-13.zip`	PA194C; same dense/native literal machine as four other C packets; alternate independent audit and structural search records.
`pa195_conpa_2026-11-13.zip`	Positive original PA195: unchanged diagonal191 program, 68−7 reader allocation, detailed ordinary PA calibration.
`pa195_diagonal_combined_2026-11-13.zip`	PA195DC combines bundled diagonal191 and7 folds with cached-head Shift/MP; 3676 sites,4784 contracts,57922 startup, six changed native rows. Optional PA195R and PA195N implement parameter-permuted and negative-vacuity emitters at195, with explicitly limited endpoint claims.

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Packet	Principal contribution or outcome
pa195_followup_research.zip	M195S8 keeps diagonal191 but pays one extra scanner control and obtains8 folds:135+68-8=195. It rejects the associated194 quotient at source ordinal1003 because seven address zeros trigger the old scanner. Includes additional equivalent source-calculus families and bounded searches.
pa196_research_2026-11-13.zip	Exact 157-node representation of 3,812-site original program; loses contextual sharing, no physical reduction.
pa208_research_2026-10-29.zip	Eighteen further helper layouts; 27 designated U208 presentations give 20 graphs, not two.
pa_below196_research.zip	Eight equivalent gates, membership/equality orientations, broad grammar searches; original201-token best 196.
pa_consistency_comparison_2026_11_13.zip	Modal-head/explosion split, twelve semantic compositions, joint gate; physical results 196–200.
pa_consistency_comparison_and_new196_2026-11-13.zip	Diagonal196 (3,700 sites), Separate196 (3,673), Scanner-best196; principal logical/source parents of positive195.
pa_consistency_cross_archive.zip	Four new196 combinations: diagonal/separate AX + Combined controller, each optionally zero-jump; sites 3,676/3,649 and startups57,922/58,168.
pa_consistency_reduction.zip	Rebracketed 196, H-only 196, integrated gate 203/206; two failed 195 diagnostics.
pa_consistency_reduction_audit_2026-11-14.zip	Four further experimental MP-H compositions, two 196-state and two 197-state; parameter/order/association/helper/scanner negative searches.
pa_consistency_research.zip	Two exact 158-node sources and repaired 196; nominal 195 needs added scanner state.
pa_consistency_research_2026-10-29.zip	Inductive 3,909→3,904 specialization; no physical sub-208.
pa_consistency_research_2026-11-13(2).zip	Twelve-rotation 196, diagonal 198, successful-MP halting-source experiment; no sub-196.
pa_consistency_research_2026-11-13.zip	Earlier diagonal-vacuity 199 and short-negation 197.
pa_consistency_research_2026-11-14.zip	NEG18: ‘INC7; INC7; node64‘, 18-site encoded negation; complete source 3,441 and 197 states (134+68−5); independent macro and scratch-zero proofs.
pa_tm_comparison_and_reductions.zip	Two rotated 196-state graphs; distributed-gate 198 with 3,773 source sites.
pa_tm_extended_research_2026-11-13.zip	Four closing-decrement variants and stronger zero invariants; all change the same row, hence not stackable.
pa_tm_research_2026-11-13(2).zip	Zerojump 196 (4,959 contracts, 58,402 startup); parity/disequality experiments; rejected 195 fails source 26.
pa_tm_research_2026-11-13.zip	Shorter-path 196 and independent physical checking; detailed direct PA reverse-translation strategy.
pa_unary_power_cover.zip	Unary weak-covering theory and PA proof translations; no new compiled small table.
u208_comparison_research_2026-10-29.zip	Original two-class audit; weak/exact schemas; rejected 207.
u208_research_2026-10-29.zip	Primary/alternative U208 and symbolic contracts; same initial history fold.